

Project Report: American Sign Language Detection

1. Objective

The objective of this project is to build a system that can detect a given American Sign Language (ASL) input image and output what the sign represents (such as letters of the alphabet).

2. Dataset Information

The dataset contains 29 classes of signs, of which 26 are for the letters A-Z and 3 classes are for SPACE, DELETE, and NOTHING. The dataset provides separate training and testing images to train a robust classification model.

3. Methodology

1. Data Preprocessing: Images were loaded, resized to 64x64 pixels, and normalized to values between 0 and 1. Data augmentation techniques like rotation, zoom, and shifting were applied to prevent overfitting.
2. Model Architecture: A Convolutional Neural Network (CNN) was implemented using TensorFlow/Keras. The model consists of multiple Conv2D, MaxPooling2D, BatchNormalization, and Dropout layers for feature extraction.
3. Model Training: The model was compiled with the Adam optimizer and categorical crossentropy loss. Early stopping was used to end training when performance stopped improving on the validation set.
4. Output: The model outputs probabilities across 29 classes.

4. Results and Conclusion

The CNN model converged successfully and demonstrates a strong capability to distinguish

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between different ASL hand gestures. The model can accurately predict the corresponding alphabet for unseen images. The trained model is saved for further inference tasks.